

Differential Oil Price Shocks and Their Effects on Brazil's Macroeconomic Performance from 1973 to 2018

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Abstract

In this paper, we examine how different types of oil price shocks, namely supply, global aggregate demand, and oil-specific demand shocks affect Brazil's macroeconomic performance from 1973 to 2018. Our paper contributes to the limited literature on differential oil shocks and their macroeconomic effects in developing countries. We use a Structural Vector Autoregression (SVAR) framework based on Kilian (2009). Cholesky decomposition using Killian's ordering is followed where we decompose the oil shocks and estimate their impacts on Brazil's GDP, inflation, and currency exchange rate with the US dollar. Our data sample is divided into two periods: the pre-2000 (Oil Import Dependent) period and the post-2000 (Post-Oil Import Dependent) Period. Our findings show that before 2000, Brazil's economy was significantly sensitive to oil, especially in terms of inflation and exchange rates. Post-2000, the effects were muted and not statistically significant. Robustness checks use alternative sample breakpoints (1995) and (2006, net exporter start point) and differing lag lengths. The robustness measure supports our findings and reveals a potential structural shift that an economy faces regarding its status going from net importer to exporter. Our findings have implications for policymakers in

resource-rich developing countries and how the effect of oil shocks changes as their status evolves from importer to exporter.

Motivation and Literature Review

Brazil's oil sector plays a large role in the country's economy. The country is Latin America's largest oil producer and the eighth largest globally, producing an average of 3.4 million barrels daily in 2023. (The Energy Research Office, 2024) In recent years, it has surpassed previous oil production records, and an analysis of data from the International Energy Agency (IEA) found that Brazil's oil production is expected to increase by 36% in the next 10 years. (350.org, 2024) These advances have been highly debated, as Brazil is set to host the UN Climate Summit, COP30, in 2025. (Climate Change News, 2025) President Luiz Inácio Lula de Silva and his administration support expanding oil exploration, including offshore exploration drilling near the mouth of the Amazon, which has raised tensions between pushing for economic growth and maintaining environmental commitments. (AP News, 2025) Furthermore, the head of Petrobras, Brazil's national oil and gas company, was fired in early 2024 by President Lula, highlighting the political pressures on Brazil's oil sector.

This dynamic, coupled with historical studies on oil price shocks and their effects on macroeconomic fluctuations, makes Brazil an interesting country to study. Brazil has transitioned from a primarily oil-importing country to a significant oil exporter over the last 20 years. Over the years, Petrobras has played a key role in boosting and advocating for national production. However, environmental and political dimensions have added more complexity to

the dynamic. The push for new exploration projects appears to conflict with previously made climate commitments. (AP News, 2025)

While many studies have looked into the macroeconomic effects of oil price shocks, the majority of this research focuses on developed countries, where data is more easily accessible. Kilian (2009) provided a fundamental framework by breaking down price shocks into three categories: supply shocks, global demand shocks, and oil-specific demand shocks. However, this framework has not been used comprehensively for developing economies, leaving the effects under-researched. Few studies have applied this framework to Brazil, and existing ones, such as Souza and de Mattos (2023), focus on a limited time frame, only considering data from 1996-2000. This does not consider important events, such as the 1973 oil crisis, the oil glut 1980s, and Brazil's 1999 currency devaluation. Because of this, many of the factors that shaped Brazil's relationship with the oil industry were not researched. There is a gap in studies regarding the effects of oil price shocks on Brazil before it became a prominent oil exporter. In this paper, we aim to answer the question: *how do different types of oil price shocks: supply shocks, global demand shocks, and oil-specific demand shocks, affect Brazil's macroeconomic performance from 1973 to 2018?*

Our hypothesis is that after Brazil became a big oil producer and exporter in 2000, this changed the country's response to global oil shocks.

Existing studies on Brazil's response to oil price shocks suggest distinct macroeconomic patterns. Oliveira et al. (2019) used a Structural Vector Autoregression, or SVAR, model to identify different macroeconomic patterns in Brazil. They found two major categories: high oil

dependency and volatility before Brazil increased oil production, and greater stability after. Similarly, Calvalcanti and Jalles (2013) used a cointegration and error correction model to examine long-term relationships between oil prices and Brazil's economic output. They found that Brazil's reliance on oil imports has decreased over the last 30 years, and more recently price shocks have little effect on output growth or inflation. Yet another paper by Oliveira et. al (2017) found that Brazil's stock market is connected to the oil market. Socially responsible firms in Brazil are still influenced by oil prices. Similarly, Mensi et al. (2017) also find co-movement between the BRICS stock market as a whole and oil prices during crises.

Although these studies offer valuable insights into Brazil's response to oil price shocks, they leave critical gaps. Existing research does not apply Kilian's structural decomposition framework to Brazil's oil price shocks over an extended time period. It may imply that important context could be missing. This paper seeks to address this by analyzing a longer time frame, 1973-2022, to assist in accounting for structural changes. Furthermore, we intend to split the data into two sections: before 2000 and after 2000 when Brazil became a large oil exporter. This will allow us to consider potential asymmetric responses to oil shocks and will improve the robustness of the model for future use forecasting. By addressing these gaps, we hope to contribute to a more nuanced understanding of the effects of oil price shocks in emerging market economies.

Data and Descriptive Statistics

We chose a period of 1973-2018 to expand previous research on Brazil's macroeconomic performance and oil shocks. Existing studies have focused primarily on 1996 onward, leaving out major oil events such as the 1973 oil crisis and the oil glut of the 1980s. Because of Brazil's

military dictatorship from 1965-1985, public data for Brazilian GDP over this extended time period is not readily accessible, however, there published research, such as that of Bonelli & Rodrigues (2012) presents a methodology for calculating quarterly and seasonally adjusted GDP for the full period. Our data comes from Oliveira et al. (2019), who used the existing methodology from Bonelli & Rodrigues with public data from the IBGE to create a public dataset with Brazilian quarterly GPA and CPI, Brazil/US exchange rate, and real oil prices. We also are using data from the Dallas Fed on the real economic activity index, and data from the EIA on crude oil production. The main limitation of our data is the lack of published data by the IBGE before 1980, causing us to rely on constructed data to get the extended time period, which may not be completely accurate. We will be using the framework from Kilian (2009) to explore how supply shocks (measured by changes in crude oil production), aggregate demand shocks (measured by the real economic activity index), and oil-specific demand shocks (measured by the real price of oil) affect Brazil's GDP, CPI, and exchange rate with the US.

Table 1: Descriptive Statistics

Variable	Mean	Median	Var	SD	Min	Max
gdp	4.6014300	4.6087652	0.1363617	0.3692719	3.7605022	5.174239
cpi	3.2822099	5.0705364	18.7372495	4.3286545	-11.5129255	5.817245
er	0.1054125	0.2056799	0.2015753	0.4489713	-0.7633551	1.050713
prod	0.1867850	-0.0313902	29.1731614	5.4012185	-23.1372347	26.691247
rea	1.2078427	-7.6026221	2948.9445439	54.3041853	-160.8394000	189.774440
rpo	23.7867661	21.6920168	111.8187386	10.5744380	6.4640357	61.587488

Table 1 shows the descriptive statistics for our 6 variables (3 shocks and 3 Brazil macroeconomic indicators). All of our variables are expressed as logs for consistent interpretation. We see from the table that GDP growth has a relatively high mean and low standard deviation, which indicates fairly stable economic growth over our period. CPI has much more variability, indicating fluctuations and periods of deflation/inflation. This makes sense, given Brazil's

historic inflation issues and attempts to curb high levels. Exchange rate changes are generally small but somewhat variable, which reflects currency fluctuations.

The oil variables are defined by Kilian (2009). Oil production has a very small mean, but a very wide range and spread which implies large amounts of volatility. Real economic activity, a measure created by Kilian, has a huge standard deviation (54.3) and is highly skewed. This means it likely is influenced by large demand shocks. The real price of oil also has large variation, which reflects global price fluctuations.

Methodology

We divided our data into two subsample periods. Oil Import Dependent Period: Pre-2000 and Post-Oil Dependent Period: Post-2000. Brazil officially became a net exporter in 2006, but to retain larger sample sizes we use 2000 as our split. This still allows us to compare the effects of oil shocks when Brazil was more dependent on oil imports vs when it held a more prominent exporting role. Separate models are estimated for each period.

First, we constructed the SVAR model on the Oil Variables: prod, rea, and rpo. This model helped us identify structural shocks and their effects on prod, rea, and rpo. The identification strategy follows a Cholesky decomposition based on the assumptions provided by Kilian (2009). The ordering of our variables was prod, rea, rpo.

$$\mathbf{u}_t = \begin{bmatrix} u_{1t} \\ u_{2t} \\ u_{3t} \end{bmatrix} = \mathbf{B} \cdot \begin{bmatrix} \varepsilon_t^{\text{supply}} \\ \varepsilon_t^{\text{agg. demand}} \\ \varepsilon_t^{\text{oil-specific demand}} \end{bmatrix} = \begin{bmatrix} b_{11} & 0 & 0 \\ b_{21} & b_{22} & 0 \\ b_{31} & b_{32} & b_{33} \end{bmatrix} \cdot \begin{bmatrix} \varepsilon_t^s \\ \varepsilon_t^d \\ \varepsilon_t^o \end{bmatrix}$$

Therefore, oil production (prod) is not affected contemporaneously by any other variable except its own shock. Aggregate demand (rea): responds contemporaneously only to oil supply and its shock. Oil prices (rpo) are affected by global supply, aggregate demand, and oil price shocks.

Then, we constructed 9 OLS models, where the exogenous variables captured oil-related shocks.

The regression equation is:

$$Y_t = \alpha + \sum_{i=1}^{12} \beta_i \cdot prod_{t-i} + \sum_{i=1}^{12} \gamma_i \cdot rea_{t-i} + \sum_{i=1}^{12} \delta_i \cdot rpo_{t-i} + \varepsilon_t$$

where:

- Y_t represents each macroeconomic variable,
- $prod_{t-i}$, rea_{t-i} , rpo_{t-i} are oil market variables and their lags, and
- ε_t is the error term.

The effects of change in crude oil production reflect supply shocks, the real economic activity index represents aggregate demand shocks, and the real oil price represents oil-specific demand shocks. The endogenous variables capture Brazilian macroeconomic indicators. These variables are Brazil's GDP (overall economic output), CPI (inflation), and Exchange Rate (currency valuation)

Results

Below we find the output and analysis of our SVAR models as well as our OLS regressions.

From these outputs, we can further investigate the response of these shocks to our macroeconomic variables and look into differences between our time periods. Throughout, we will refer to the results of Kilian (2009) to compare our results with a developing economy to those of a developed economy.

Oil Market Responses to Structural Shocks: Pre vs Post-2000

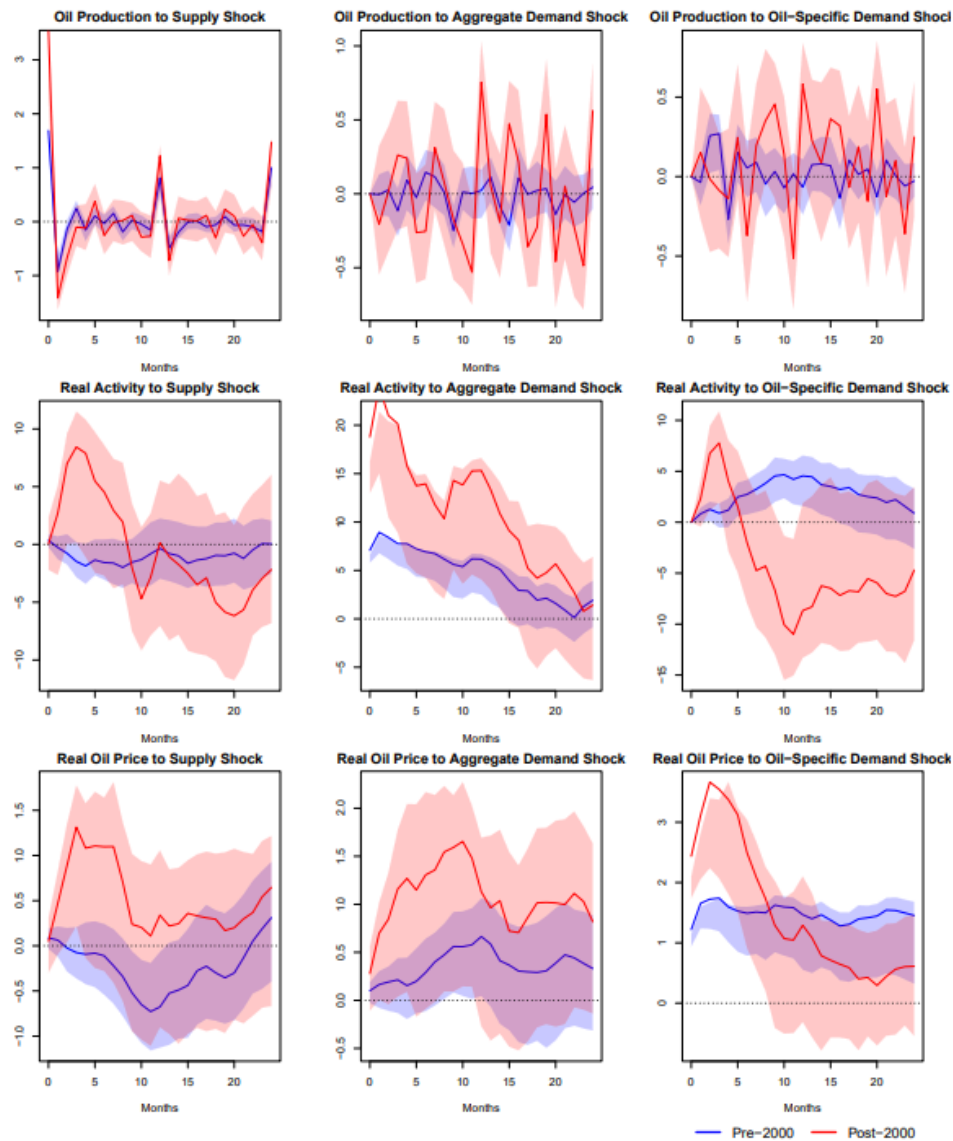


Figure 1: Impulse Response Functions of Oil Market Variables to Structural Shocks (Pre vs. Post-2000) This figure compares the dynamic responses of oil production, real economic activity, and real oil prices to structural shocks (supply, aggregate demand, and oil-specific demand) in two periods: Pre-2000 (blue) and Post-2000 (red). The results are based on a VAR(12) model estimated separately for each period using monthly data from 1973–1999 and 2000–2018. The shaded regions represent 90% confidence intervals for each period.

Supply Shocks

Supply shocks represent an unanticipated oil supply disruption, as seen in the left column. In the top left, oil production falls sharply in both periods before returning to baseline, with a small positive blip around the 12-month mark. The drop is slightly more extreme post-2000, possibly due to a growing industry, where a supply disruption has a stronger effect. This aligns with Kilian (2009), who found that supply disruptions cause a sharp decline in production followed by partial recovery. Real economic activity (middle-left) shows a slight, insignificant decline pre-2000, but a post-2000 pattern of an initial positive spike followed by a more sustained drop. The positive spike is unexpected and may result from the assumptions of the Choleski decomposition or shock misattribution. The shorter post-2000 period may also increase standard errors, making responses noisier. For rpo (bottom-left), pre-2000 shows a small initial decrease followed by a short increase, while post-2000 displays a more volatile and sustained rise. This differs from Kilian's finding that supply shocks have only modest, short-lived effects on oil prices, suggesting possible changes in market structure or policy effects over time.

Aggregate Demand Shocks

The center column shows responses to aggregate demand shocks. Oil production (top-center) remains close to zero in both periods, with some post-2000 volatility. This is expected, as production responds slowly to demand due to supply chain and investment lags. The post-2000 increase in variation may be driven by emerging industrial economies. Real economic activity (middle-center) shows a small, significant increase pre-2000 and a larger, more persistent increase post-2000, consistent with Kilian's (2009) finding that demand shocks have strong and lasting effects. The sharper post-2000 response suggests heightened sensitivity to global demand. The real price of oil (bottom-center) also increased slightly pre-2000, but much more strongly

and persistently post-2000, reflecting increased demand pressure, consistent with Kilian's findings and likely intensified by financialization and emerging market growth.

Oil-Specific Demand Shocks

The right column shows responses to oil-specific demand shocks. Oil production (top-right) is largely unresponsive and is insignificant in both periods, aligning with Kilian (2009), who noted that production does not react to future scarcity expectations. Real economic activity (middle-right) shows a small increase pre-2000, but post-2000 shows a positive spike followed by a sustained decline—deviating from Kilian's findings. This drop may reflect greater sensitivity to volatility or uncertainty in the later period. Rpo (bottom-right) rises in both periods, with a longer significant increase pre-2000 and a briefer, but still sharp, increase post-2000. This reflects precautionary behavior, as future supply concerns drive prices up, which is consistent with Kilian's interpretation of oil-specific demand shocks.

Table 2: FEVD: Pre- and Post- 2000

Pre-2000	Oil Production			Real Economic Activity			Real Price of Oil		
	prod	rea	rpo	prod	rea	rpo	prod	rea	rpo
Month 0	100.0	0.0	0.0	0.1	99.9	0.0	6.8	1.0	92.2
Month 6	81.3	12.7	6.0	0.9	98.2	0.9	3.1	17.7	79.2
Month 12	79.4	16.4	4.2	0.8	95.9	3.3	5.0	23.1	71.9
Month 18	74.1	18.7	7.2	1.3	91.3	7.4	6.7	23.3	69.9
Month 24	76.2	17.7	6.1	1.6	87.1	11.3	9.3	20.5	70.2

Post-2000	Oil Production			Real Economic Activity			Real Price of Oil		
	prod	rea	rpo	prod	rea	rpo	prod	rea	rpo
Month 0	100.0	0.0	0.0	3.6	96.4	0.0	3.5	3.4	93.1
Month 6	80.3	18.1	1.6	16.2	76.2	7.7	7.4	52.4	40.2
Month 12	75.9	17.8	6.3	22.7	73.5	3.8	33.4	54.9	11.7
Month 18	74.0	19.1	6.8	18.5	79.0	2.5	22.6	68.9	8.5
Month 24	74.5	18.2	7.3	18.9	78.6	2.5	23.0	70.8	6.2

The above tables show the Forecast Error Variance Decomposition of each oil shock. There are separate tables for pre- and post- 2000, and the variables are shown as headers. The shocks are represented along the columns, and the FEVDs for months 0, 6, 12, 18, and 24 are printed.

The FEVD tables offer insight into how each shock explains the variation in each of the oil variables over two years. We are using our subsamples to compare the FEVD before and after 2000. Pre-2000, we see that oil production (as shown on the left column) is dominated by its own shocks, as it starts at 100% in month 0 and is still at 76.2% after 24 months. Real economic activity and real price of oil shocks play only small roles. In the post-2000 period, we see very similar results; however, the role of production has slightly decreased, and the roles of real economic activity and real price of oil shocks have slightly increased in explaining the variation of oil production.

Real economic activity is also almost exclusively explained by its own shocks over both time periods. In the pre-2000 period, real price of oil shocks explained the majority not already explained by rea, but in the post-2000 period, most of this additional variation is explained by production. This indicates an increased sensitivity to production shocks in the post-2000 period.

The right column, the real price of oil, has much differentiation between the sample periods. Pre-2000 shows much of the variation being explained by its own shocks; however, by 1 year that variation is down to 71.9%. Most of this lost variation is picked up by rea shocks, which account for over 20% from month 12 to month 24. In the post-2000 period, only 11.7% variation in rpo is explained by its own shocks after 12 months, and that is down to only 6.2% by 24 months. Real economic activity shocks pick up the majority of this, and production shocks explain close to 25% by 24 months.

Overall, these results reveal increased interdependence among oil variables in the post-2000 period. Particularly, the real price of oil becomes increasingly influenced by other forces (supply and aggregate demand). This suggests a shift away from precautionary pricing.

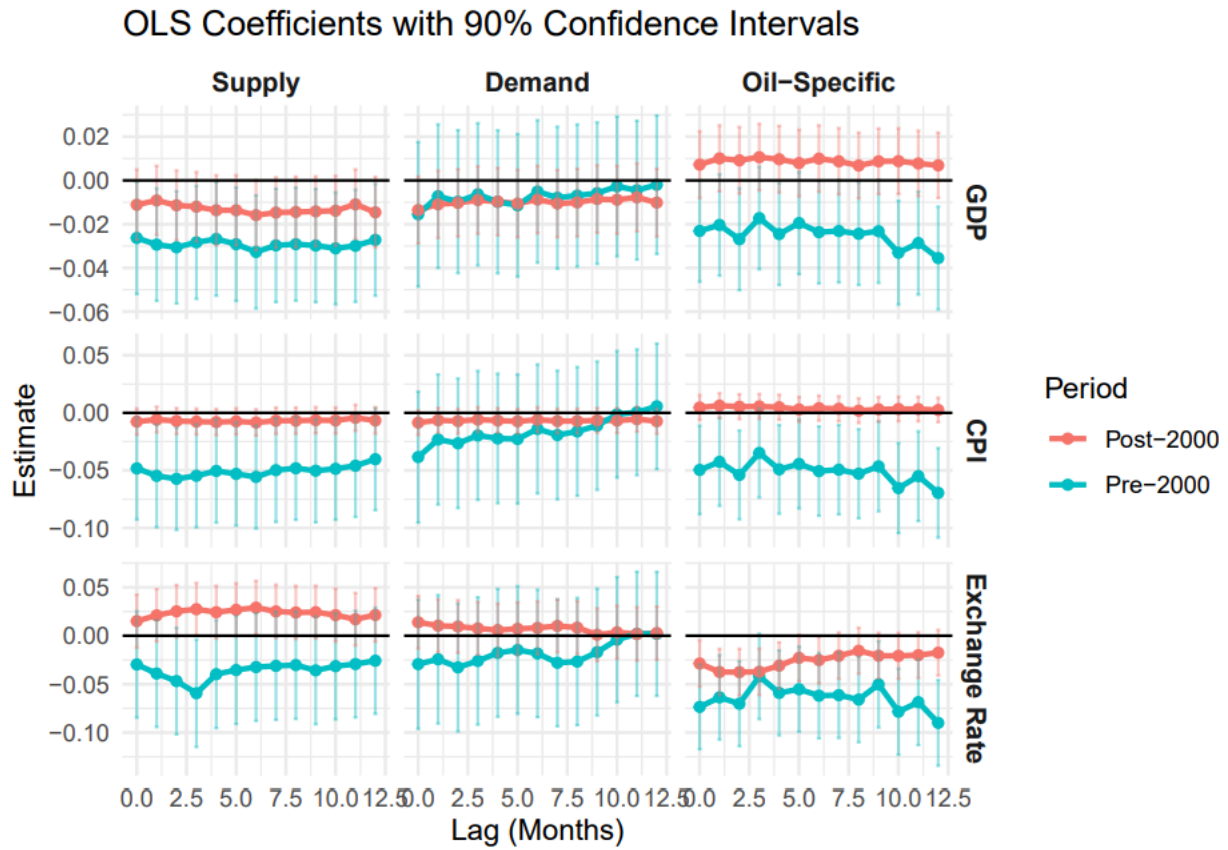


Figure 2: OLS Coefficients of Oil Shock Impacts on Brazil’s GDP, CPI, and Exchange Rate with 90% Confidence Intervals (Pre- and Post-2000). This figure displays the estimated OLS coefficients for the effect of supply, aggregate demand, and oil-specific demand shocks on our three Brazilian macroeconomic variables across 12 lags. Each panel includes responses for both the pre-2000 period (blue) and post-2000 period (red). The black horizontal lines are at an estimate of 0. Overall, we see that the pre-2000 responses seem to have stronger effects across all variables.

Supply Shocks

In response to oil supply shocks, we observe larger and more persistent effects in the pre-2000 period across all three variables. Brazil’s CPI fell significantly and remains fairly steady, which suggests that inflation in Brazil was more sensitive to oil disruptions during this time. This makes sense, as Brazil was experiencing weaker monetary policy and greater dependence on oil

imports. The exchange rate decreases for about 3 months before beginning to slowly recover, and GDP shows a small negative trend. In the post-2000 period, responses are much more muted and statistically insignificant, which implies improved resilience to these shocks, possibly due to structural reforms and Brazil's growing oil exports. This suggests a shift from vulnerability to more stability from oil supply volatility. In developed countries like the U.S., Kilian (2009) finds only minor CPI effects from supply shocks. Our stronger pre-2000 responses likely reflect Brazil's less flexible price systems at the time and suggest a slightly different response between developed and developing countries.

Aggregate Demand Shocks

Aggregate demand shocks have relatively mild, and statistically insignificant effects on Brazil's economy in the post-200 period, but pre-2000 seems to have initial slight negative effects followed by a consistent return to the original levels for all 3 variables. These results suggest that aggregate demand shocks do not have strong short-term effects on Brazil's GDP, CPI, or exchange rate. This could be due to the increased implementation of economic stabilization and inflation targeting post-2000, which could have lessened the reactions.

Oil-Specific Demand Shocks

Much like with production and aggregate demand shocks, oil-specific demand shocks also tend to have stronger effects in the pre-2000 time period, while the post-2000 responses tend to be relatively stable and insignificant. GDP and CPI experience decline in the pre-period, aligning with economic theory, where precautionary demand raises prices and lowers output. The muted effects post-2000 reflect greater economic stability and reduced vulnerability to global oil shocks. The exchange rate had weak responses in both periods, however, increased stability

post-2000 indicates more credible monetary and fiscal policy and can be attributed to Brazil's inflation targeting starting in 1999 and the Fiscal Responsibility Law in 2000 (Bonomo, 2024).

Robustness Check

Subsample Analysis

Table 3: Significant Coefficient Lags in the OLS regressions over different Subsamples

Variable	Shock	Pre-1995	Post-1995	Pre-2000	Post-2000	Pre-2006	Post-2006
GDP	Supply	NA	L6	L0-12	NA	L0, 4-12	NA
GDP	Aggregate Demand	NA	NA	NA	NA	NA	L0, 5-10, 12
GDP	Oil-Specific	L0-2, 4, 7-12	NA	L2, 4, 6, 8, 10-12	NA	L7-8, 10-12	NA
CPI	Supply	NA	NA	L0-11	NA	L0-12	NA
CPI	Aggregate Demand	NA	NA	NA	NA	NA	L7-10, 12
CPI	Oil-Specific	L0-12	NA	L0-2, 4-12	NA	L1-12	NA
ER	Supply	NA	L4-9	L3	L3-6	L6-10	NA
ER	Aggregate Demand	NA	NA	NA	NA	NA	NA
ER	Oil-Specific	L0-12	L1-4	L0-2, 4-12	L0-4, 6	L0-12	L0-7, 10-11

To test the sensitivity of our results to the choice of 2000 as the structural break date, we re-estimated the OLS models using alternative breakpoints at 1995 and 2006, each splitting the full sample into pre- and post-periods starting in 1973 and ending in 2018. Table 3 above highlights the splits and significant lags during each period. Across the 1995 and 2000 splits, our main results remain roughly consistent, particularly with oil-specific demand shocks in the pre-period yielding several statistically significant lags. We do see that in the 1995 split, a few lags of supply shocks are significant, which is not seen across either the 2000 or 2006 split. This

could be due to shortening the pre-split period, reflecting the historical volatility and Brazil's reliance on oil imports.

Additionally, when we change the breakpoint to 2006 (when Brazil became an oil exporter), we observe a difference: the aggregate demand shock in the post-2006 period exhibits statistically significant lagged effects not present in earlier subsample specifications. This suggests that the structure of the oil market and its responsiveness to aggregate demand shocks may have changed beginning around 2006. The remainder of our results remained consistent in this time period.

We interpret this as a structural shift in global oil demand dynamics. This period from 2006-2018 captures the global commodity boom and increased industrial demand from many emerging Asian markets. These together likely increased the persistence of aggregate demand shocks. This suggests that the 2000-2006 period likely damped this effect, making the results not significant with the break at 2000. This is likely due to the recession in 2001 and slow recovery after the .com bubble burst. Having this in the same period as the high volatility and growth of 2006-2018 would likely dampen the effects.

Despite these new findings post-2006, we retain 2000 as our breaking point, as it allows better balance and interpretability. The longer period provides a more stable window for inference, and the break at 2000, as 2000 marks a key shift in global economic integration and the rise of emerging market demand. Although we do not adjust our splitting point, we report the 2006 results to highlight potential heterogeneity, and it is important to keep in mind for potential future studies, as using 2006 could be more beneficial with a lengthened time frame.

Lag Length for the VAR models

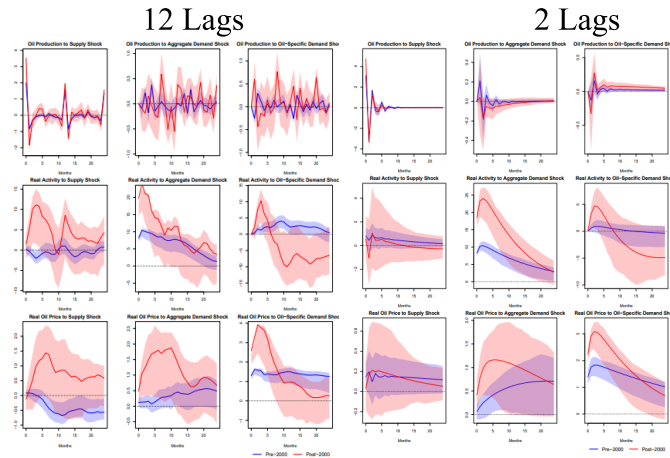


Figure 3: IRFs across different lag lengths. This graph shows the impulse response functions from our global oil variables over 12 lags (AIC optimized) and 2 lags (BIC optimized)

Table 4: Lag Length Robustness			
Shock	Pre	Consistency	Post_Conistency
Supply		64	68
Aggregate Demand		88	88
Oil-Specific		24	96

To check the robustness of our lag length selection, we ran 2 VAR models: one with 12 lags as we used in our analysis (optimal under AIC), and one with 2 lags (optimal under BIC/SC). Table 4 shows the consistency (in %) between the 2 models. Consistency here is measured by the percent of IRF lags with the same significance status across both models. As seen in Figure 3, the general reactions seem to be the same, with less noise/variability in the model with only 2 lags. Supply shocks have moderately consistent results in both periods, which suggests some sensitivity to lag length. Aggregate demand shocks prove to be highly consistent, with close to 90% of the lags having the same consistency status. Oil-specific shocks are very inconsistent before 2000 but perfectly consistent post-2000, which suggests much greater sensitivity in the earlier periods. Overall, our VAR results tend to be more robust post-2000 and highlight the importance of testing alternative specifications.

Implications

Our results have several implications for economic policy for developing countries like Brazil.

The asymmetry in Brazil's macroeconomic responses before and after becoming a major oil producer supports our hypothesis that a shift from net oil importer to exporter changes how oil shocks transmit through the economy. This suggests applying frameworks of differentiating oil shocks like those provided by Kilian (2009) majorly used to examine developed countries without controlling for such structural differences may lead to incomplete conclusions in developing country contexts

The stronger macroeconomic responses to oil shocks in the pre-2000 period, especially inflation and exchange rate volatility highlight the vulnerability of Brazil's economy to external shocks. These findings reinforce the importance of measures like inflation targeting starting in 1999 (BCB) and fiscal responsibility law in 2000 which may have strengthened Brazil's response to shocks after the post-2000 period. Policymakers in other resource-rich countries with structurally weak economies may draw on how Brazil designed macroeconomic frameworks to stabilize oil-related volatility (Afonso et al., 2016)

Lastly, our robustness checks indicate that the specific timing of Brazil's structural shift (2000 vs 2006) may affect the estimated magnitude of shock responses. This finding suggests the need for future studies to experiment with alternative breakpoints.

Limitations

One of the main limitations is the short window of official Brazilian macroeconomic data available in quarterly units. Since our time period spans earlier than official IBGE data, we are

relying on constructed data, which, despite a reputable methodology, may not be entirely accurate and less preferable than official data.

Another potential limitation is the different time periods between our Brazilian variables and the global oil variables. Since our oil variables are monthly and Brazil's ones are quarterly, we are not able to run a full VAR to examine their relationship. Aggregating the monthly variables would cause us to lose crucial information, as those tend to be quite volatile on the monthly scale. To combat this, we followed Kilian's method and ran OLS regressions. We flat-filled the GDP so each set of 3 months had the corresponding GDP value. This is a limitation, as it oversimplifies the true GDP/CPI/Exchange Rate monthly variation.

Finally, we are relying on the methodology of Kilian (2009) for our motivation and methodology, and we are following his assumptions for the Choleski decomposition. Although his methods are widely accepted, they impose strong assumptions on order, which may not truly represent global dynamics.

Conclusion

Our paper examined the differential oil shocks and their effect on the macroeconomic indicators of Brazil between 1973 and 2018. We applied Kilian's (2009) structural VAR model and Cholesky decomposition assumptions. The analysis was separated into two time periods (pre vs post-2000). Our findings show that Brazil's macroeconomic variables had a higher sensitivity to oil shocks, particularly inflation and exchange rate, pre-2000. This effect is likely due to its reliance on oil imports and weaker institutions. In contrast, the post-2000 period shows statistically insignificant macroeconomic responses, suggesting Brazil's growing role as an oil exporter, combined with stronger monetary and fiscal frameworks compared to before. Our

paper contributes to the limited literature on differential oil price shocks in developing countries by extending the time frame from 1973-2018. It also highlights the importance of accounting for structural changes like increasing oil production when evaluating oil shock responses. While our findings are specific to Brazil, they may provide lessons for other resource-rich developing countries that are undergoing structural shifts like expanding their oil sectors.

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